The Hearing-Aid Speech Quality Index (HASQI) Version 2

JAMES M. KATES, *AES Fellow,* **AND KATHRYN H. AREHART**
(James.Kates@colorado.edu) (Kathryn.Arehart@colorado.edu)

University of Colorado, Boulder, CO 80309, USA

This paper presents an improved index for predicting speech quality. The Hearing-Aid Speech Quality Index (HASQI) version 2 is based on a model of the auditory periphery that incorporates changes due to hearing loss. The auditory model outputs are used to measure changes in the signal envelope, temporal fine structure, and spectrum caused by the hearing aid or audio system processing. The index comprises a nonlinear term sensitive to noise and nonlinear distortion and a linear term sensitive to long-term spectral changes, and the quality prediction is formed by taking the product of these two terms. The nonlinear term combines measurements of envelope and temporal fine structure modifications, and it is shown to lead to more accurate predictions than using nonlinear terms that are based on envelope or temporal fine structure measurements alone. Compared to the previous version of HASQI, the new index offers improved accuracy for a variety of degraded signal conditions including frequency compression, noise suppression, speech replaced with the output of a noise vocoder, acoustic feedback and feedback cancellation, and speech combined with modulated noise.

0 INTRODUCTION

Signal degradations, such as additive noise or nonlinear distortion, can reduce speech quality. In audio and telecommunication systems, it is generally assumed that any change in the signal corresponds to a loss of quality. Hearing aids present a more complicated situation since most of these devices contain nonlinear processing that is intended to make portions of the speech more audible to the hearing-impaired user, and the "degraded" signal may be modified on purpose to enhance audibility. A quality index for use with hearing aids must therefore predict the trade-offs between signal distortion and audibility for hearing-impaired listeners. This paper presents a speech quality index that considers modifications of the speech signal in the context of the normal or impaired auditory periphery. The index predictions are compared to quality ratings obtained from hearing-impaired as well as from normal-hearing listeners for a variety of signal-processing conditions, including nonlinear hearing-aid processing.

Many degradation mechanisms simultaneously affect the signal envelope and the signal temporal fine structure (TFS). Additive noise, for example, reduces the envelope modulation depth by filling in the pauses in the speech and also corrupts the TFS of the speech by adding jitter corresponding to the random fluctuations of the noise. Peak clipping reduces the signal modulation depth by removing the signal peaks and also modifies the TFS by introducing additional frequency components corresponding to the harmonic distortion products. Thus for many forms of signal degradation changes to the signal envelope and to the TFS are closely related, and quality can be accurately predicted by measuring the changes to either quantity.

Changes to the signal TFS are often measured by computing the cross-correlation between the unmodified and degraded signals. In one speech quality index (Tan et al. [1]; Tan and Moore [2]), the signal is passed through an auditory filter bank and then divided into 30-ms segments. The cross-correlation between the reference and degraded signals is computed for each segment and normalized by the energy in the segments. A level weighting function is applied to reduce the importance of low-intensity segments, and the normalized and weighted cross-correlations are then averaged within each frequency band.

The coherence can also be used to compute a quality index. Coherence is computed by taking the cross-correlation between a noise-free unprocessed reference signal and the noisy processed signal and dividing by the product of the root-mean-squared (RMS) intensities of the two signals (Carter et al. [3]; Kates [4]). The magnitude-squared coherence can be converted into a signal-to-distortion ratio (SDR), which can then be used in a manner similar to the signal-to-noise ratio (SNR) to estimate the signal degradation. Kates et al. [5] formulated a speech quality index that combined the SDR values computed for low-, mid-, and high-intensity segments of the speech in a manner similar

to the coherence SII (CSII) intelligibility index (Kates and Arehart [6]).

An alternative to using the TFS as the basis for predicting speech quality is to use the signal envelope. The HASQI quality index (Kates and Arehart [7]) uses the envelope outputs produced in each frequency band of an auditory model. The output envelopes are smoothed and divided into segments. At each segment interval, the short-time spectrum is approximated by fitting a set of half-cosine basis functions to the envelope outputs sampled across the frequency bands. The result is a set of spectral components that vary over time. The quality index is computed using the normalized cross-correlations of the spectral components of the degraded speech with those of the reference speech. In the PEMO-Q index (Huber and Kollmeier [8]), the signal envelope is extracted from each frequency band in an auditory model. The envelope in each frequency band is passed through a modulation filter bank. The degraded signal is compared to the clean reference signal by computing the normalized cross-correlation of each modulation filter output at each auditory analysis frequency and taking the average.

Although accurate quality indices have been based on both the signal cross-correlation and on the signal envelope, there are some situations where an index based exclusively on one or the other may fail. An example where the envelope may be insufficient is the noise vocoder (Dudley [9]; Shannon et al. [10]), in which the speech is replaced by bands of noise having the same envelope modulation as the speech. The change to the signal envelope is small, but replacing the speech with modulated noise can cause substantial reductions in quality (Anderson et al. [11]). Vocoded speech is relevant to hearing aids since techniques related to noise or tone vocoding have been proposed for decorrelating the high-frequency outputs of a hearing aid to improve the effectiveness of feedback cancellation processing (Ma et al. [12]; Kates [13]).

An example of where coherence may be insufficient is ideal binary mask noise suppression (Kjems et al. [14]; Ng et al. [15]), where segments of noisy speech within a frequency band are suppressed if the local signal-to-noise ratio falls below a threshold. Reducing the amplitude of noisy speech segments will not affect the short-time coherence between the clean speech and noisy speech computed over that segment. However, the annoyance due to the noisy segment may be reduced since the potential for the reduced-amplitude segment to be masked by an adjacent segment has been increased. Noise suppression is found in many hearing aids, so it is important to accurately predict the effectiveness of this type of processing.

These weaknesses in using only the signal envelope or coherence in predicting speech quality suggest that an index that measures both envelope and TFS changes in the signal will be more accurate when considered over a range of signal modifications. The objective of the present paper is to describe a new quality index that combines envelope and TFS measurements to produce a model that is more robust than the original version of HASQI. The design goals for the new index include computational efficiency and accuracy for both normal-hearing and hearing-impaired listeners. The new index, HASQI version 2, uses an auditory model that incorporates aspects of normal and impaired peripheral auditory function (Kates [16]). The cepstral correlation is computed from the envelope output in each frequency band and provides a measurement of the fidelity with which the envelope time-frequency modulation has been preserved. The vibration correlation is computed from the basilar-membrane vibration output in each frequency band of the auditory model and provides a measurement sensitive to the changes in the speech temporal fine structure.

The remainder of the paper starts with a description of the noise and distortion data used to train the original version of HASQI and the new HASQI version 2. The quality index is validated using several independent datasets, including quality ratings for frequency compression for speech in a background of multi-talker babble, noise suppression, hearing aids driven into acoustic feedback, speech where the high frequencies are replaced by the output of a noise vocoder, and speech with additive modulated noise. An overview of these published datasets is presented next. The auditory model used for the new index is then described, followed by a description of how the envelope, basilar-membrane vibration, and internal spectrum outputs of the auditory model are combined to produce HASQI version 2. Modeling results are then presented for the training and validation datasets, and the new model results are also compared to results from other quality models that have been proposed.

1 QUALITY RATING DATA

The quality index was trained using the same data as was used for the original version of HASQI (Arehart et al. [17]; Kates and Arehart [7]), thus facilitating the comparison of the new index with the previous one. An additional reason for training the index on just one dataset is the difficulty of combining quality ratings across experiments. Each experiment generates its own set of perceptual anchors, depending on the type and amount of signal modification present in the stimuli (Yeung and Soman [18]). Thus similar stimuli in different experiments could be assigned different quality ratings depending on the range and type of other stimuli present.

After training, the new index was tested against data from several other quality rating experiments. These experiments include frequency compression (Souza et al. [19]), ideal binary mask noise suppression (Arehart et al. [20]), hearing aids with varying amounts of acoustic feedback and feedback cancellation (Bondy et al. [21]), speech with modified temporal fine structure produced by a noise vocoder (Anderson et al. [11]), and speech with additive modulated noise (Jin et al. [22]). The original stimuli from the training and the five validation experiments were used.

1.1 Training Data

The participants in the Arehart et al. [17] subjective experiments were a group of 14 adults with normal hearing

and a group of 15 adults with mild to moderate-severe hearing losses. The test materials were two sets of concatenated sentences from the hearing-in-noise test (HINT) (Nilsson et al. [23]), with one concatenated two-sentence set spoken by a male talker and another two-sentence set spoken by a female talker (Nilsson et al. [24]). The stimuli were digitized at a 44.1 kHz sampling rate and down sampled to 22.05 kHz to give a bandwidth similar to that found in commercial hearing aids (Kates [25]). The task of the listener was to indicate the rating of sound quality on a rating scale that ranged from 1 (poor sound quality) to 5 (excellent sound quality) (ITU-R BS.1284-1 [26]). For computing the quality indices, the ratings for each subject averaged over the stimulus repetitions were normalized so that the highest observed score was reset to 1 and the lowest score was reset to 0. The rating normalization reduced the intersubject variability caused by different subjects adopting different internal anchors or using only part of the rating scale.

The processing conditions were implemented using a simulated hearing aid programmed in MATLAB. The order of processing was additive noise, followed by nonlinear processing, which was followed by the linear filters. The final processing step was to adjust the loudness of the filtered signal to match that of the unprocessed reference. The level of presentation for the subjects in the normal-hearing (NH) group was 72 dB SPL, and the stimuli were amplified for listeners in the hearing-impaired (HI) group using the NAL-R prescriptive formula based on individual thresholds (Byrne and Dillon [27]). Stimuli were presented to the listeners monaurally using headphones in a sound booth.

The pair of male stimuli and the pair of female stimuli were subjected to processing conditions comprising noise and nonlinear distortion, linear filtering, and combinations of noise and distortion with linear filtering. In addition, an unprocessed "clean speech" reference condition was included in the stimulus set.

The noise conditions included speech in stationary speech-shaped noise and speech in multi-talker babble at a range of signal-to-noise ratios (SNRs). The distortion conditions included instantaneous peak clipping and amplitude quantization. Dynamic-range compression (Kates and Arehart [28]) was included both for the clean speech and for speech in babble. Spectral subtraction (Tsoukalis et al. [29]) was included to give a set of noise-suppression conditions for speech in babble at different SNRs. The final noise and nonlinear condition combined spectral subtraction with compression for speech in babble.

The linear processing conditions represented spectral changes typically found in hearing aids. The linear filters included high-pass and low-pass filters having a range of cutoff frequencies. In addition, combined low-pass and high-pass filters were used to give a set of band-pass conditions. Wideband spectral changes were implemented using both positive and negative spectral tilts. The linear conditions also included individual resonance peaks and multiple resonance peaks representative of hearing-aid acoustic and feedback resonances.

The combined processing conditions comprised all possible combinations of six nonlinear conditions with six linear conditions. The noise and nonlinear conditions comprised stationary noise, peak clipping, amplitude quantization, compression, and babble plus compression. The linear conditions were high-pass and low-pass filters, positive and negative spectral tilts, a single resonance peak, and a set of three resonance peaks.

1.2 Validation Data

1.2.1 Frequency Compression

Frequency compression (Aguilera Muñoz et al. [30]; Simpson et al. [31]; Glista et al. [32]) is intended to improve the audibility of high-frequency speech sounds by shifting them to lower frequency regions where listeners with high-frequency hearing loss have better hearing thresholds. The data used in this paper comprise the quality ratings for frequency-compressed speech reported by Souza et al. [19]. Fourteen adult listeners with normal to near-normal hearing and 26 adult listeners with mild-to-moderate high-frequency loss participated in the experiments.

The stimuli for the quality ratings comprised a two-sentence set spoken by a female talker (Nilsson et al. [24]). The stimuli were digitized at a 44.1 kHz sampling rate and down-sampled to 22.05 kHz. The task of the listener was to indicate the rating of overall sound quality on a 10-point rating scale. The sentences were used in quiet and in multi-talker babble at SNRs ranging from 10 to –10 dB in steps of 5 dB. For this paper, only the quality ratings for the positive SNRs and no noise were used to give a stimulus range compatible with the training dataset. Quality ratings at poorer SNRs are dominated by the loss of intelligibility, so a different model would potentially be needed for speech quality at negative SNRs. After the addition of the babble, the sentences were processed using frequency compression.

Frequency compression was implemented using sinusoidal modeling (McAulay and Quatieri [33]). In the present implementation, the signal was divided into low-frequency and high-frequency bands. The low-frequency signal was used without further modification and sinusoidal modeling was applied to the high-frequency signal. The ten highest peaks in the high-frequency band were selected, and the amplitude and phase of each peak were preserved while the frequencies were reassigned to lower values. Output sinusoids were then synthesized at the shifted frequencies (Quatieri and McAulay [34]; Aguilera Muñoz et al. [30]) and combined with the original low-frequency signal to produce the frequency-compressed output. The frequency-compression parameters represented the range that might be available in wearable hearing aids and included three frequency compression ratios (1.5:1, 2:1, and 3:1) and three frequency compression cutoff frequencies (1, 1.5, and 2 kHz). A control condition having no frequency compression was also included.

Following frequency compression, the speech signals were amplified for the individual hearing loss using the National Acoustics Laboratories-Revised (NAL-R) linear prescriptive formula (Byrne and Dillon [27]). The sentences were presented monaurally to the listeners through headphones. The stimulus level prior to NAL-R amplification

was 65 dB SPL. For evaluating the quality indices, the ratings for each subject averaged over the stimulus repetitions were normalized so that the highest observed score was reset to 1 and the lowest score was reset to 0.

1.2.2 Noise Suppression

Arehart et al. [20] obtained quality ratings for noisy speech processed through an ideal binary mask noise-suppression algorithm. The data presented in this paper were obtained from thirty older subjects having mild to moderate hearing losses and from seven younger subjects having normal hearing. The test materials were two sentences spoken by a female talker from the IEEE corpus (Rosenthal [35]). All stimuli were digitized at 44.1 kHz and were down-sampled to 20 kHz for the noise-suppression processing. The background noise was four-talker babble at signal-to-noise ratios of –6 to +12 dB. The sentence level prior to noise suppression was set to 65 dB SPL.

The noisy speech stimuli were processed with a binary mask noise-reduction strategy (Kjems et al. [14]; Ng et al. [15]). The target speech signal, the masker signal, and the target-plus-masker mixture were each separated into frequency bands using an analysis filterbank consisting of 64 gammatone filters (Patterson et al. [36]) with center frequencies equally distributed on the equivalent rectangular bandwidth (ERB) scale (Moore and Glasberg [37]) over the frequency range of 50 and 8000 Hz. The processing was done in time frames each having a duration of 20 ms with 50-percent overlap, resulting in a new frame every 10 ms.

A time-frequency cell consists of one frame in one frequency band. In each time-frequency cell, the local signal-to-noise ratio was determined from the separate clean target and noise masker signals. The local SNR was then compared to a local criterion (LC) of 0 dB, resulting in an ideal binary mask decision equal to 1 if the local SNR was above LC and 0 otherwise. The data of Kjems et al. [14] indicate that an LC of 0 dB is most effective for SNRs in the range of approximately +5 to –10 dB. Similar to the procedure in Li and Loizou [38], errors were introduced into the ideal binary mask by randomly flipping a certain percentage (0, 10, and 30 percent) of the time-frequency units either from 0 to 1 or from 1 to 0. The binary patterns were then converted into gain values, where 1's were converted into 0 dB gain and the zeros were converted into an attenuation of either 10 dB or 100 dB. The noisy speech signal was then multiplied by the binary gain values to give the processed signal in the frequency domain. The processed signal was then filtered through a time-reversed gammatone filterbank, thereby ensuring the same processing delay at all frequencies to provide accurate reconstruction of the signal transients.

Following the noise suppression processing, the speech signals were amplified for the individual hearing loss, when present, using the National Acoustics Laboratories-Revised (NAL-R) linear prescriptive formula (Byrne and Dillon [27]). Stimuli were presented monaurally over headphones. The speech quality was rated on a 10-point scale, with 0 representing the worst quality presented and 10 representing the highest quality. For evaluating the quality indices, the ratings for each subject averaged over the stimulus repetitions were normalized so that the highest observed score was reset to 1 and the lowest score was reset to 0.

1.2.3 Noise Vocoder

Anderson et al. [11] obtained quality ratings for speech where the high frequencies of noisy speech were replaced by the output of a noise vocoder. Ten subjects with normal hearing and ten subjects with hearing loss participated in the experiments. The test materials were two sentences spoken by a female talker from the IEEE corpus (Rosenthal [35]). All stimuli were digitized at 44.1 kHz and were down-sampled to 22.05 kHz. The background noise was multi-talker babble.

The speech was processed without any noise and at SNRs of 18 and 12 dB. The pair of noisy sentences was passed through a bank of 32 band-pass filters having center frequencies distributed on a standard equivalent rectangular bandwidth (ERB_N) filter scale (Slaney [39]). The signal envelope for each vocoded band was generated via the Hilbert transform. The Gaussian noise used for the noise vocoding was passed through the same filterbank as the speech. Two vocoded signals were produced. One signal was produced by multiplying the filtered noise by the speech envelope. For the second signal, the noise envelope fluctuations were first removed by dividing the filtered noise by its own envelope before multiplying it by the speech envelope. As a last processing step, both the speech and noise were passed through the same filters as in the first filtering stage to remove any out-of-band modulation products, and the RMS level of the vocoded signal in each frequency band was matched to that of the original speech.

Quality was rated by the listeners as the number of frequency bands subjected to noise vocoding was varied at each SNR. The amount of noise vocoding was increased in steps of two frequency bands from no bands vocoded to the 16 highest-frequency bands vocoded. The cutoff frequency of the 16-band vocoded condition was 1.6 kHz, so in this condition all bands above 1.6 kHz were vocoded and all bands below were the original speech. The subjects judged the overall quality of the sentences using a scale ranging from 0 to 10 (Gabrielsson et al. [40]). The stimulus level for the normal-hearing listeners was 65 dB SPL, and NAL-R amplification (Byrne and Dillon [27]) was provided for the hearing-impaired listeners. The stimuli were presented monaurally over headphones. For evaluating the quality indices, the ratings for each subject averaged over the stimulus repetitions were normalized so that the highest observed score was reset to 1 and the lowest score was reset to 0.

1.2.4 Feedback Cancellation

Bondy et al. [21] obtained speech quality ratings for sentences processed through hearing aids with varying amounts of acoustic feedback and feedback cancellation. Nineteen normal-hearing and thirteen hearing-impaired

subjects took part in the experiments. The test material was a single sentence spoken by a male talker.

Hearing aids from several manufacturers were mounted on the KEMAR manikin (Burkhard and Sachs [41]). The test sentence was reproduced over a loudspeaker in a test booth. The gain of each hearing aid without feedback cancellation was adjusted so that the instrument was just below, at, and above the maximum stable gain, which is defined as the gain that just caused audible acoustic feedback ("whistling") in the device. The hearing-aid output was then recorded for later playback. The recording procedure was repeated with the feedback cancellation processing turned on. A total of 60 different recorded sentences were obtained.

The pre-recorded stimuli were presented to the subjects at a sampling rate of 24.4 kHz diotically through Audio-Technica ATH-M40fs circumaural headphones. Subjects adjusted the intensity of a token presentation to give their most-comfortable listening level and then rated their preference (like-dislike) for the 60 test sentences on a 9-point scale (Peryam and Pilgrim [42]). For evaluating the quality indices, the ratings for each subject averaged over the stimulus repetitions were normalized so that the highest observed score was reset to 1 and the lowest score was reset to 0.

1.2.5 Speech-Modulated Noise

Jin et al. [22] tested ten subjects with normal hearing. The test material comprised a concatenated pair of sentences spoken by a male talker extracted from the hearing-in-noise test (HINT) (Nilsson et al. [23]). The stimuli included one pair of sentences without any noise and eleven noisy sentences. The sampling rate was 22.05 kHz. Subjects were tested monaurally using headphones.

The stimuli were processed through a 16-band filterbank covering the frequencies from 80 Hz to 10 kHz. The speech envelope was extracted in each frequency band using the Hilbert transform. Stationary Gaussian noise was passed through the same filter bank as the speech. Fully-modulated noise was formed by multiplying the noise in each frequency by the filtered speech envelope, while unmodulated noise bypassed the envelope multiplication. Intermediate amounts of noise modulation were produced by mixing the stationary noise with the fully-modulated noise while holding the overall long-term noise level constant. The amount of noise modulation varied from 0 to 100 percent, and SNR values of 0 to 20 dB were used along with speech not having any added noise.

Listeners rated the stimuli on three scales: overall preference, distortion, and noisiness using a paired-comparison procedure. Only the overall preference data are used in this paper since these data are closest to the quality ratings of Arehart et al. [17] used to train the quality index; the distortion and noisiness scales give decreasing scores as quality increases. The subjects rated the different SNRs in different test blocks, so stimuli at one SNR were never compared to stimuli at a different SNR. Each SNR thus has a different set of perceptual anchors, and the results for one SNR cannot be directly compared to the results for another. Instead, the correlation coefficient between the index and the subject ratings was computed for each of the three SNRs, and the three values averaged to produce an overall correlation coefficient. For evaluating the quality indices, the ratings for each subject averaged over the stimulus repetitions at each SNR were normalized so that the highest observed score was reset to 1 and the lowest score was reset to 0.

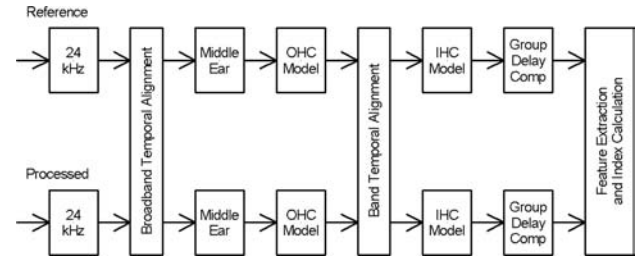

Fig. 1. Block diagram showing the reference and processed signal comparison.

2 AUDITORY MODEL

The approach to predicting speech quality used in HASQI version 2 is to compare the output of an auditory model for a degraded test signal with the output for an unprocessed input signal. A detailed description of the new auditory model is presented in Kates [16] and is summarized here. The model is an extension of the Kates and Arehart [7] auditory model; that model has been shown to give envelope outputs that can be used to produce accurate predictions of speech quality for normal-hearing and hearing-impaired listeners under a wide variety of noise, nonlinear distortion, and linear filtering conditions.

The overall model block diagram is presented in Fig. 1. The comparison of the processed and reference signals requires that they be temporally aligned, so part of the model is the temporal alignment of the signals. The first alignment step is a rough alignment of the broadband signals that removes large delay differences. Each signal then goes through the middle ear and cochlear mechanics models. A second temporal alignment step then removes any remaining timing differences between the reference and processed signals in each frequency band by adjusting the signal delay to maximize the cross-correlation of the signals in each band. The separate signals then go through the inner hair-cell (IHC) model. The last temporal alignment step is across frequency for each signal rather than between the reference and processed signals and compensates for the frequency-dependent group delay of the auditory filterbank. In the final processing step the auditory model outputs are converted into the signal features for comparing the processed signal with the reference signal.

The basilar membrane (BM) vibration signal in each frequency band is compressed using the same control function as for the envelope in that band, so the envelope of the BM vibration tracks the envelope output. The auditory threshold for the BM vibration signal is represented as a low-level

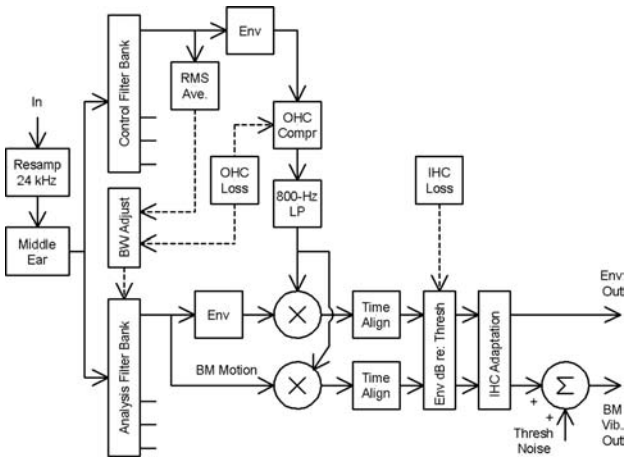

Fig. 2. Block diagram of the auditory model used to extract the signals in each frequency band.

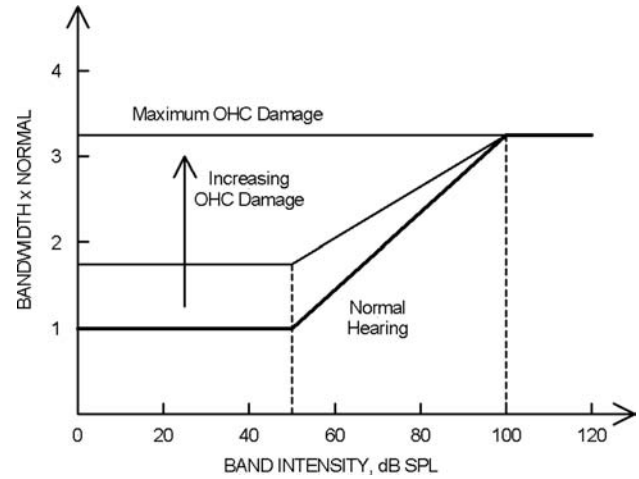

Fig. 3. Example of the adjustment of the auditory filter bandwidth with increasing signal intensity.

additive white noise. Both the envelope and the vibration outputs are available for modeling speech quality.

The reference signal for the quality index is the clean signal combined with NAL-R equalization (Byrne and Dillon [27]) for the hearing loss and passed through a model of the impaired auditory periphery. The use of NAL-R equalization for the reference is based on the results of Arehart et al. [17], in which the NAL-R filtered signal was found to yield the highest quality ratings for the hearing-impaired listeners. The degraded signal also includes the amplification to compensate for the hearing loss and is also passed through the model of the impaired periphery.

The processing for one signal is shown in the block diagram of Fig. 2. The auditory model starts with sample rate conversion to 24 kHz, followed by the middle ear filter. The next stage is a linear auditory filter bank, with the filter bandwidths adjusted to reflect the input signal intensity and the increase in filter bandwidth due to outer hair-cell (OHC) damage. Dynamic-range compression is then provided, with the compression controlled by a separate control filter bank. The amount of compression is reduced with increasing OHC damage. Hearing loss due to IHC damage is represented as a subsequent attenuation stage, and IHC firing-rate adaptation is also included in the model. The envelope output in each frequency band comprises the compressed envelope signal after conversion to dB above auditory threshold; the BM vibration signal is centered at the carrier frequency for each band and is modified by the same amplitude modulation as the envelope signal.

Several changes have been made relative to the auditory model used by Kates and Arehart [7]. First is setting the sampling rate to 24 kHz for all signals. Sampling-rate conversion is provided in the model for signals at higher or lower rates. The shape of the infinite-impulse-response (IIR) filters used for the middle ear and auditory filter banks depends on the sampling rate. Using a fixed sampling rate ensures that the magnitude transfer functions and group delays in the model remain the same independent of the original sampling rate of the signals and that the ultimate model predictions are independent of the input signal sampling rate.

Second is the addition of an intensity adjustment in the auditory filter bandwidth. The shape of the auditory filters depends on the intensity of the input signal as well as on the degree of hearing loss, with the filters becoming broader as the signal intensity increases (Ruggero et al. [43]; Baker and Rosen [44]; Baker and Rosen [45]). The change in bandwidth is shown schematically in Fig. 3. For normal hearing, the filter bandwidth is set to the ERB (Moore and Glasberg [37]) for intensities below 50 dB SPL. For impaired hearing, the bandwidth at and below 50 dB SPL is set to the bandwidth computed for the amount of OHC damage related to the hearing loss. For both normal and impaired hearing, the bandwidth is set to the bandwidth corresponding to maximum OHC damage for intensities at or above 100 dB SPL. Linear interpolation of the filter bandwidth is used for intensities between 50 and 100 dB SPL. In addition, the range of auditory filter center frequencies has been increased to cover 80 to 8000 Hz, as opposed to the 150 to 8000 Hz used by Kates and Arehart [7].

The outer hair cell (OHC) dynamic-range compression has also been adjusted to better reflect the consensus in the literature. The compression rule for normal hearing is modeled by three line segments as shown by the bold lines in Fig. 4. Inputs within 30 dB of normal auditory threshold (0 dB SPL) receive linear gain. Inputs between 30 and 100 dB SPL are compressed. The system reverts to linear gain for inputs above 100 dB SPL. The compression ratio in the model for normal hearing increases linearly with ERB number from a compression ratio of 1.25:1 at 80 Hz to a compression ratio of 3.5:1 at 8 kHz. This compression behavior is consistent with physiological measurements of compression in the cochlea (Cooper and Rhode [46]) and with psychophysical estimates of compression in the human ear (Hicks and Bacon [47]; Plack and Oxenham [48]).

OHC damage shifts the auditory threshold and reduces the compression ratio, as shown by the thin lines in Fig. 4. The dependence of the compression behavior on OHC

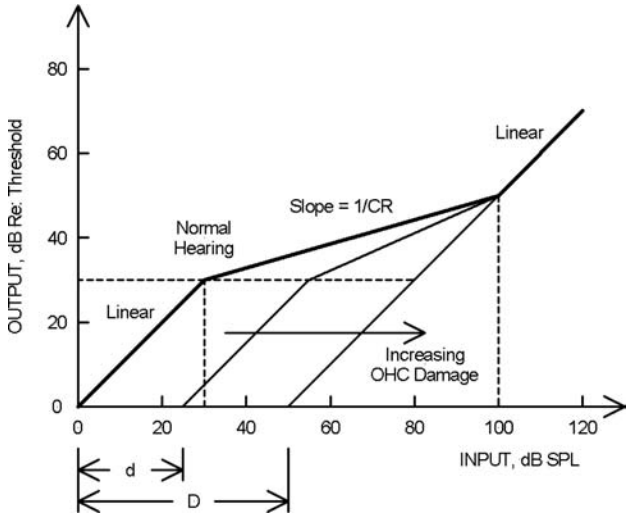

Fig. 4. Compression input–output behavior as a function of OHC damage. The bold line shows the compression for normal ear and the thin lines show the compression for increasing amounts of OHC damage. The maximum amount of OHC damage gives a linear system.

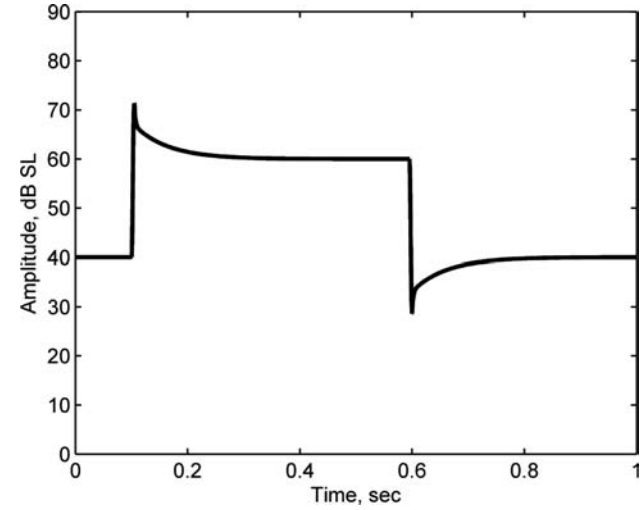

Fig. 5. Rapid and short-time IHC adaptation for an increase of 20 dB in the signal level.

damage reproduces the loudness recruitment found in hearing-impaired listeners (Kiessling [49]). The shifted curves are constructed so that an input of 100 dB SPL in a given frequency band always produces the same output level independent of the amount of OHC damage. The maximum loss due to OHC damage is indicated by *D* in the figure, and for this degree of loss the system is reduced to linear amplification. Intermediate amounts of OHC damage result in an intermediate shift of the compression behavior, as indicated by the loss given by *d* in the figure.

The final changes to the model are the addition of adaptation of the inner hair-cell (IHC) neural firing rate and the addition of compensation for the auditory filter group delay. The IHC synapse provides the rapid and short-term adaptation observed in the neural firing rate (Harris and Dallos [50]; Gorga and Abbas [51]). The rapid adaptation time constant is 2 ms and the short-term time constant is 60 ms. An example of the synapse response is shown in Fig. 5. The compressed signal envelope in the frequency band is initially at 40 dB SL. The level jumps to 60 dB SL at 100 ms and returns to 40 dB SL at 600 ms. Raised-cosine 5 ms windows are applied to both level changes, which reduces the overshoot in comparison with instantaneous transitions. Compensation for the group delay of the gammatone filters is then applied to the output of the IHC synapse model since internal compensation for the filter delay appears to occur higher in the auditory pathway (Wojtczak et al. [52]).

3 QUALITY INDEX COMPONENTS

The speech quality indices examined in this paper compare the auditory model outputs for a reference signal to the outputs for a processed (degraded) signal. The noise and nonlinear index compares the patterns of envelope time-frequency modulation and/or temporal fine structure while ignoring any long-term spectral changes. The linear filtering index compares the long-term spectral representations of the two signals while ignoring the short-term differences in signal modulation and temporal fine structure. Unlike the previous index of Kates and Arehart [7], which used separate models for the NH and HI listener groups, the new model parameters are the result of a joint optimization over the combined NH and HI listeners.

The quality index has very few free parameters. For the nonlinear portion of the model, various powers of the cepstral correlation (Kates and Arehart [7]) and vibration correlation (described below) were tested, and raising these terms to the third power was found to give the minimum root mean-squared (RMS) error when compared to the training data. Polynomial fits resulted in the third-order term having a weight of 1 and the remaining terms having weights of zero. The combination of cepstral correlation squared times the vibration correlation also yielded the smallest RMS error of the different products and sums of powers tested. For the linear portion of the model, a linear combination of the standard deviations of the spectral difference and spectral slope difference (Tan et al. [1]; Kates and Arehart [7]) was chosen, and the weights selected to give the minimum RMS error fit with the linear training data. The index was fit only to the training data described in Sec. 1.1; the datasets described in Sec. 1.2 were used only for validation of the resultant model.

3.1 Cepstral Correlation

One candidate for measuring the nonlinear degradation of the signal is cepstral correlation. The calculation is similar to the procedure used by Kates and Arehart [7] in computing the HASQI index. The envelope samples output by the auditory model, when taken across frequency at a given time slot, constitute a short-time log magnitude spectrum on an auditory frequency scale. The inverse Fourier transform of this log spectrum produces a set of coefficients

that are similar to the mel cepstrum (Imai [53]). The cepstrum computation is performed in the frequency domain by fitting the auditory model envelope outputs with a set of half-cosine basis functions. These basis functions are very similar to the principal components for the short-time spectra of speech (Zahorian and Rothenberg [54]) and have been used for accurate machine recognition of both consonants (Nossair and Zahorian [55]) and vowels (Zahorian and Jagharghi [56]). The basis functions are given by:

$$b_j(k) = \cos[(j-1)\pi k/(K-1)], \tag{1}$$

where j is the basis function number and k is the gammatone filter index for frequency bands 0 though K-1 for $K = 32$.

Let $e_k(m)$ denote the sequence of smoothed sub-sampled envelope samples in frequency band k for the reference signal, and let $d_k(m)$ be the envelope samples for the degraded signal. The envelope smoothing is provided by 16-ms von Hann windows having 50-percent overlap, giving a smoothed envelope sampling rate of 125 Hz. The reference-signal cepstral sequence $p_j(m)$ and the degraded-signal sequence $q_j(m)$ are then given by:

$$\begin{aligned} p_j(m) &= \sum_{k=0}^{K-1} b_j(k) e_k(m) \\ q_j(m) &= \sum_{k=0}^{K-1} b_j(k) d_k(m) \end{aligned} \tag{2}$$

The cepstrum correlation is computed by taking the cross-correlation of the cepstral sequences for the reference and degraded signals. The sequence values corresponding to pauses in the speech are removed from cepstral sequences $p_j(m)$ and $q_j(m)$, and the mean value is subtracted from each edited sequence to yield the zero-mean edited sequences $\hat{p_j}(m)$ and $\hat{q_j}(m)$. The pauses were deleted by converting the envelopes in each band to linear values to give numbers related to specific loudness (Moore and Glasberg [57]; Kates [16]), summing the values across frequency and then converting the sum back to dB. The segments less than 2.5 dB re:threshold were removed from the correlation calculation. The normalized covariance is then given by:

$$r(j) = \frac{\sum\limits_{m \in Speech} \hat{p}_j(m)\hat{q}_j(m)}{\left[\sum\limits_{m \in Speech} \hat{p}_j^2(m)\right]^{1/2} \left[\sum\limits_{m \in Speech} \hat{q}_j^2(m)\right]^{1/2}}. \tag{3}$$

The average cepstrum correlation given by the average of the normalized covariance values $r(2)$ though $r(6)$:

$$c = \frac{1}{5}\sum_{j=2}^{6} r(j). \tag{4}$$

The nonlinear model using just the cepstral correlation is a MMSE third-order regression fit to the combined NH and HI subject ratings for the noise and distortion stimuli. The nonlinear model based on the cepstral correlation alone is given by:

$$Q_{Nonlin} = c^3. \tag{5}$$

A monotonic model was chosen to allow for consistent extrapolation to conditions worse than those used in the training dataset.

3.2 Vibration Correlation

A second candidate for measuring the nonlinear degradation of the signal is the normalized cross-correlation of the BM vibration in each auditory band. The calculation uses the BM vibration outputs of the auditory model in each frequency band and is motivated by the work of Tan et al. [1] and Tan and Moore [2]. The BM vibration signal in each band is divided into 16-ms segments having a 50-percent overlap, with each segment windowed using a von Hann window. The mean of the segments is removed, and each windowed segment of the degraded signal is cross-correlated with the corresponding segment of the reference signal. The cross-correlation is normalized by the reference and degraded signal magnitudes to give a value corresponding to the short-time coherence within the segment.

Let $x_k(n)$ be the BM vibration for the reference signal and $y_k(n)$ be the BM vibration for the degraded signal in frequency band k. The signals after being windowed and converted to zero-mean are given by $\hat{x}_k(n)$ and $\hat{y}_k(n)$. The normalized cross correlation for segment m in frequency band k is given by:

$$z(m,k) = \underset{\tau}{Max}\left\{\frac{\sum\limits_n \hat{x}_k(n)\hat{y}_k(n+\tau)}{\left[\sum\limits_n \hat{x}_k^2(n)\right]^{1/2}\left[\sum\limits_n \hat{y}_k^2(n)\right]^{1/2}}\right\}, \tag{6}$$

where the delay τ is chosen to yield the maximum value of the cross-correlation over the range of time lags from 1 to –1 ms.

Each normalized cross-correlation value is multiplied by a frequency-dependent weight $w(m, k)$ that is set to 0 if the segment of the reference speech lies below auditory threshold and is set to the IHC synchronization index for segments above threshold. The synchronization index gives the degree to which the neural firing pattern is synchronized to the temporal fluctuations of the signal within the band and decreases with increasing frequency (Johnson [58]; Palmer and Russell [59]). The synchronization index as a function of frequency was set to a fifth-order lowpass filter with a cutoff frequency of 3.5 kHz (Heinz et al. [60]); this filter was found to yield the most accurate quality predictions. The weighted cross-correlations are summed across segment and frequency band and are divided by the sum of the weights to give the BM vibration index:

$$v = \frac{\Sigma_k \Sigma_m w(m,k) z(m,k)}{\Sigma_k \Sigma_m w(m,k)}. \tag{7}$$

The nonlinear model using just the vibration correlation is a MMSE third-order regression fit to the combined NH and HI subject ratings for the noise and distortion stimuli. The nonlinear model based on the vibration correlation alone is given by:

$$Q_{Nonlin} = v^3. \tag{8}$$

A monotonic model was again chosen to allow for consistent extrapolation to conditions worse than those used in the training dataset.

3.3 Cepstral × Vibration Product

The third nonlinear model considered is the combination of the cepstral correlation and BM vibration correlation models. The most accurate combination of the first- and second-order terms and cross products was found to be the product of the square of the cepstral correlation index times the BM vibration index. The product index is given by:

$$Q_{Nonlin} = c^2 v. \tag{9}$$

The product index is monotonic in each of its constituent components.

3.4 Spectral Shape

The linear index for predicting speech quality is based on changes to the long-term signal spectrum. Linear filters and spectral modifications all affect the long-term spectrum of the signal but have essentially no effect on the envelope correlations between the reference and filtered signals that are measured by the nonlinear model. The linear model for measuring the long-term spectral effects is the same approach as used by Kates and Arehart [7] and which is based in turn on the linear model of Moore and Tan [61].

Moore and Tan [61] have proposed a model of sound quality for spectral modifications that uses differences in excitation patterns and the differences in the slopes of the excitation patterns. The excitation pattern is an internal representation of the spectrum of the stimulus after the cochlear filters and neural transduction. Their model is used as the basis for the spectral difference model implemented in this paper, but with the auditory model of this paper substituted for their excitation pattern calculation. The output of the system is the average level in each analysis band converted to dB re:threshold. Prior to the calculations the levels are converted back to linear amplitude, and the input and output spectra are normalized to both have RMS values of 1 when summed across the complete set of auditory bands. The level normalization removes the signal amplitude as a factor in the linear model, leaving only the spectral differences as factors.

Let $|\hat{X}(k)|$ be the normalized input linear signal spectrum and $|\hat{Y}(k)|$ be the normalized output signal spectrum. The difference in the spectra is given by:

$$d_1(k) = \left|\hat{Y}(k)\right| - \left|\hat{X}(k)\right|, 0 = \mathrm{k} = \mathrm{K} - 1 \tag{10}$$

and the difference in the spectral slopes is given by:

$$d_2(k) = \left[\left|\hat{Y}(k)\right| - \left|\hat{Y}(k-1)\right|\right] - \left[\left|\hat{X}(k)\right| - \left|\hat{X}(k-1)\right|\right] \quad 1 \le k \le K-1 \tag{11}$$

The standard deviation of the spectral difference is then:

$$\sigma_1 = g\left\{\frac{1}{K}\sum_{k=0}^{K-1}\left[d_1(k) - \bar{d}_1\right]^2\right\}^{1/2}, \tag{12}$$

where g is a scaling factor empirically set to 0.4, and the standard deviation of the slope difference is given by:

$$\sigma_2 = \left\{\frac{1}{K-1}\sum_{k=1}^{K-1}\left[d_2(k) - \bar{d}_2\right]^2\right\}^{1/2}, \tag{13}$$

where the overbar in both equations denotes the average over the frequency bands. Both standard deviations have a minimum value of 0 and were limited to have a maximum value of 1.

The linear model is a linear combination of the spectrum and slope standard deviations. Standard deviations of zero for both the spectral difference and spectral slope difference would indicate a perfect match between the processed and reference signal spectra, which would correspond to a perfect quality rating. Therefore, the model output is adjusted to start at 1 (no loss of quality) and is reduced as the standard deviations of the spectrum and slope increase. The model is a MMSE linear regression fit to the combined NH and HI subject ratings for the linear filtered stimuli. The linear model is given by:

$$Q_{Linear} = 1 - 0.579\sigma_1 - 0.421\sigma_2. \tag{14}$$

A monotonic model was again chosen to allow for consistent extrapolation to conditions worse than those used in the training dataset.

3.5 Combined Index

The HASQI version 2 metric is a combination of the index derived for noise and nonlinear processing with the index derived for linear processing. A multiplicative model (Kates and Arehart [7]) was found to be the most accurate in reproducing the subject ratings for the combined stimuli. The multiplicative model is given by:

$$Q_{Combined} = Q_{Nonlin} \times Q_{Linear}. \tag{15}$$

In the absence of noise and distortion the multiplicative model is identically the linear index, and in the absence of linear filtering the multiplicative model is identically the noise and nonlinear processing index. The multiplicative model has no free parameters and is monotonic in its two constituent components.

4 RESULTS

Table 1 presents the results for the complete model using each of the three candidate nonlinear terms. The BM vibration correlation nonlinear term corresponds to a model that uses only the temporal fine structure, the cepstral correlation term corresponds to a model that uses only the envelope, and the c^2v nonlinear term uses the combination that was found to give the highest prediction accuracy. Each of these terms was combined with the linear model given by Eq. (14) using the multiplicative procedure given by Eq. (15). The results for the original version of HASQI are also included in the table. The entries in the table are the Pearson correlation coefficients measured for the indicated combinations of subject group and dataset. The predictions and subject ratings were averaged over the subjects

Table 1. Pearson correlation coefficients for candidate models for the normal-hearing (NH), hearing-impaired (HI), and combined NH and HI subject groups. The models were fit to the combined NH plus HI subject groups.

		Version 2 Nonlinear Term			
Dataset	Subject Group	v^3	c^3	c^2v	HASQI Ver 1
Noise and	NH only	.940	.960	.959	.965
Distortion	HI only	.933	.964	.965	.934
	NH & HI	.952	.973	.974	.964
Frequency	NH only	.922	.969	.979	.936
Compress	HI only	.870	.978	.974	.968
	NH & HI	.900	.978	.981	.960
Noise	NH only	.526	.932	.891	.819
Suppress	HI only	.726	.986	.954	.969
	NH & HI	.636	.975	.937	.929
Noise	NH only	.956	.888	.926	.866
Vocoder	HI only	.921	.933	.953	.864
	NH & HI	.945	.916	.946	.871
Feedback	NH only	.906	.617	.795	.792
Cancel	HI only	.848	.894	.889	.889
	NH & HI	.922	.915	.930	.898
Modulated Noise	NH only	.939	.992	.987	.993
Average	NH only	.865	.893	.921	.895
	HI only	.860	.951	.947	.925
	NH & HI	.877	.955	.957	.931

in each group before computing the correlation coefficients for the processing conditions. All correlation coefficients have $p < 0.001$. The correlation coefficients were computed separately for each subject group, and the two groups were given equal weight when computing the correlation coefficients over the combined subject groups. The average was computed giving each dataset equal weight; since the modulated noise data were available only for normal-hearing listeners, the correlation coefficient for this dataset was given a weight of 0.5 when computing the average over the combined NH and HI subject groups.

All of the proposed models, as well as the previous version, did well with the noise and distortion training dataset. Note that the correlation coefficients are for the complete set of processing conditions, which comprises noise and nonlinear distortion without any long-term spectral changes, linear filtering without any distortion, and conditions combining noise and nonlinear distortion with linear filtering. The old version, despite being fit to just the HI listener group, does not do as well for the HI listeners as the new version using the cepstral correlation cubed (c^3) (Eq. (5)) or the cepstral correlation squared times the vibration correlation (c^2v) (Eq. (9)) nonlinear terms. The new version using the vibration correlation (v^3) (Eq. (8)) nonlinear term is not as accurate for either subject group as the implementations that include the cepstral correlation.

For the frequency compression dataset, the two new versions using cepstral correlation are more accurate than the new version using v^3. The old version of HASQI has accuracy comparable to that for the new c^3 and c^2v versions for the HI listeners but is not as accurate for the NH listeners. Overall, for the combined NH and HI subjects, the new c^3 and c^2v versions offer the best performance.

For the noise suppression dataset, the new c^3 implementation gives the highest accuracy for both subject groups and for the average over the groups. The v^3 model is substantially worse than the others. The new c^2v model gives performance that is better than the old HASQI for the NH listeners but not as good for the HI listeners. When averaged over the subject groups, the new c^3 model gives the highest accuracy and the v^3 model gives the lowest.

A different model ranking emerges for the noise vocoder dataset. For these processing conditions the envelope was mostly preserved but the temporal fine structure was modified. The v^3 model gives the highest accuracy for the NH listeners, while the c^2v model gives the highest accuracy for the HI listeners. Averaged over the combined subject groups, these two models give comparable accuracy. The new c^3 model and the old version of HASQI give the worst accuracy for these data.

The feedback cancellation dataset represents processing conditions that were substantially worse than those used to train the models, so these data test the models when extrapolated to very poor quality. The new v^3 model gives the highest accuracy for the NH subjects while the new c^3 model gives the highest accuracy for the HI subjects. The accuracy of the old HASQI and the new c^2v models are comparable for the NH and HI subject groups, but the new c^2v model gives the best performance for the combined subject groups.

For the modulated noise conditions, only NH subjects were tested in the experiments. The new v^3 model gives accuracy that is noticeably lower than for the other three models. The accuracy of the new c^3, c^2v, and the old HASQI models are similar.

The results shown in Table 1 indicate that no one modeling approach is superior for all conditions. In general, when evaluating a device such as a hearing aid, there will be several forms of noise, nonlinear distortion, and linear filtering present, and one does not know a priori which

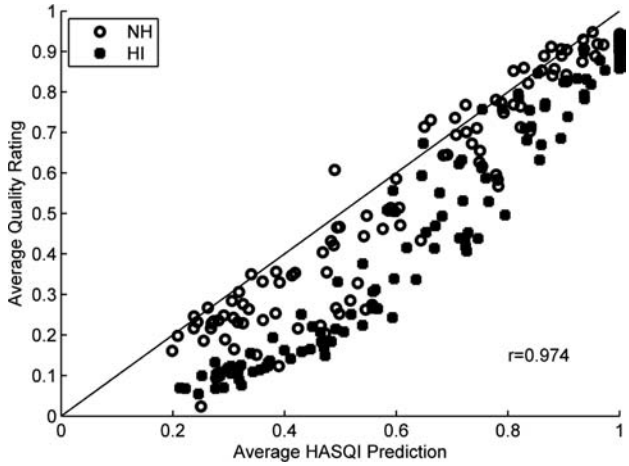

Fig. 6. Quality predictions for the noise and distortion data, averaged over the combined normal-hearing and hearing-impaired subject groups.

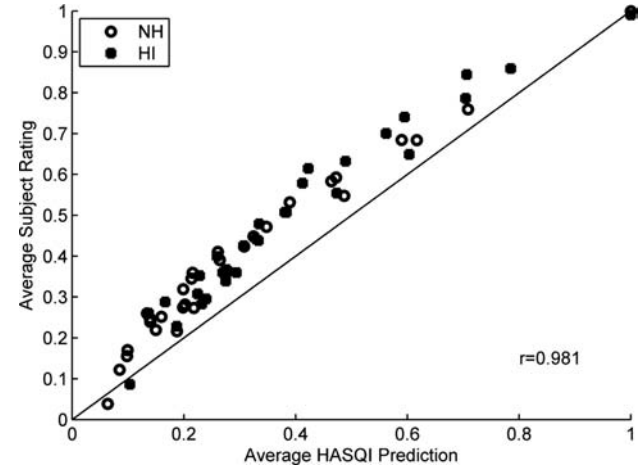

Fig. 7. Quality predictions for the frequency compression data, averaged over the combined normal-hearing and hearing-impaired subject groups.

quality model would be the best choice. A reasonable criterion for selecting the "best" model is therefore the average over all of the datasets presented in the table. The new c^2v model gives the highest average correlation coefficient for the NH listeners, nearly the highest for the HI listeners, and the highest for the combined NH plus HI subject groups. The c^2v model also has the advantage of never having the worst performance for any combination of processing and subject group and having the best performance for the combined NH plus HI groups for four out of the six datasets. The c^2v model is therefore the preferred approach and offers superior average performance in comparison to the old version of HASQI.

Scatter plots for the c^2v version of HASQI are presented in Figs. 6–10 for the first five datasets presented in Table 1. The x-axis in each figure is the new HASQI prediction, and the y-axis is the subject ratings. The open circles are for the NH listeners and the closed squares are for the HI listeners. Each point represents one processing condition averaged over the subjects in the corresponding hearing group. For points above the diagonal line the model predicts lower quality than the subject ratings, while for points below the diagonal line the model predicts higher quality. Note that each of the datasets represents a different set of subjects and different signal processing, resulting in different perceptual anchors for each experiment. Furthermore, the quality ratings for each subject in each dataset were normalized to use the entire 0 – 1 scale, so the highest quality rating in any given experiment is not necessarily perfect reproduction and may not match the best quality available in a different experiment. The differences in perceptual anchors and normalization mean that the slopes of the fit of the predicted to observed ratings do not necessarily match those of the training data and will differ from one dataset to the next.

The scatter plot for the noise and distortion dataset is presented in Fig. 6. The highest quality in this experiment was the unprocessed stimuli, so the highest predicted quality is 1 since the unprocessed signal is compared to itself. However, the highest subject quality rating for this condition is less than 1 given the variability in the subject ratings. As indicated by the correlation coefficients in Table 1, the ratings for both the NH and HI listeners are reproduced accurately. However, the predicted ratings for the HI listeners lie below the diagonal line and are thus higher than the observed ratings. One possible explanation would be that the auditory model assumes that more of the reference and degraded signals lie below the impaired auditory threshold than is actually the case, so audible noise and distortion are erroneously assumed to be inaudible.

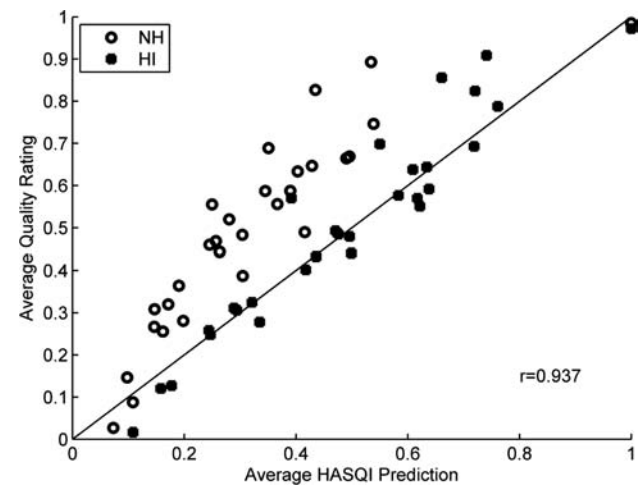

Fig. 8. Quality predictions for the noise suppression data, averaged over the combined normal-hearing and hearing-impaired subject groups.

The scatter plot for the frequency compression dataset is presented in Fig. 7. The predictions for the NH and HI listeners are similar and are slightly lower than those given by the subjects. The noise suppression dataset predictions are plotted in Fig. 8. The predictions for the NH listeners are lower than the observed ratings, while the predictions for the HI listeners are very close to the observed ratings. The lower ratings for the NH listeners indicates

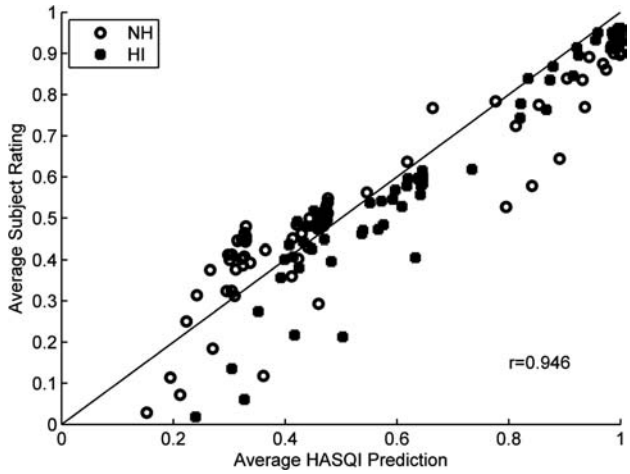

Fig. 9. Quality predictions for the noise vocoder data, averaged over the combined normal-hearing and hearing-impaired subject groups.

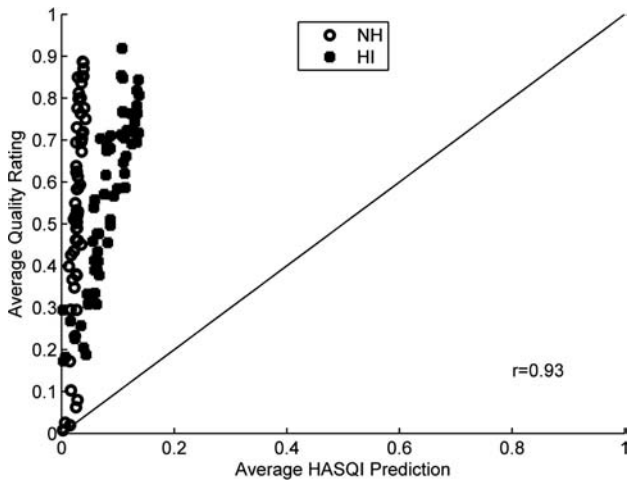

Fig. 10. Quality predictions for the feedback cancellation data, averaged over the combined normal-hearing and hearing-impaired subject groups.

that the model is more sensitive to the distortion introduced by the frequency compression than are the NH subjects. The noise vocoder dataset predictions are presented in Fig. 9. The variability of the observed ratings and the accuracy of the predictions is similar for both the NH and HI groups.

The scatter plot for the feedback cancellation dataset is presented in Fig. 10. Unlike the experiments documented in the previous four figures, the feedback cancellation experiment did not include an undistorted signal as part of the stimulus set. Also, the stimulus playback to the subjects was at a listener-selected comfortable level rather than at a level set by the experimenter, so the audibility of the signal and of the noise and distortion components may also vary from one subject to the next. Despite these limitations, the new HASQI model does a good job of predicting the relative quality ratings. The points are all above the diagonal because the subject ratings were normalized to give a maximum of 1. The rating normalization had the advantage of reducing the intersubject variability associated with each subject's perceptual anchors, but also forced the best condition in the experiment to have a rating of 1 even though an undistorted reference signal was not provided to the listeners. The steep slopes of the model predictions are a direct result of this rating normalization because degraded signals in this experiment were assigned the same normalized quality ratings as the undistorted signals provided in the other listening experiments used for the index validation. The points for the NH and HI groups of listeners appear to follow different regression lines, although the range for the NH and HI listeners is similar due to the rating normalization. The steeper slope for the NH relative to that for the HI listeners indicates that HASQI is predicting higher quality ratings for the HI group. These better predictions may be the result of the model assuming that more of the distortion products or "whistling" in the hearing aids is below audibility for HI subjects than for the NH subjects.

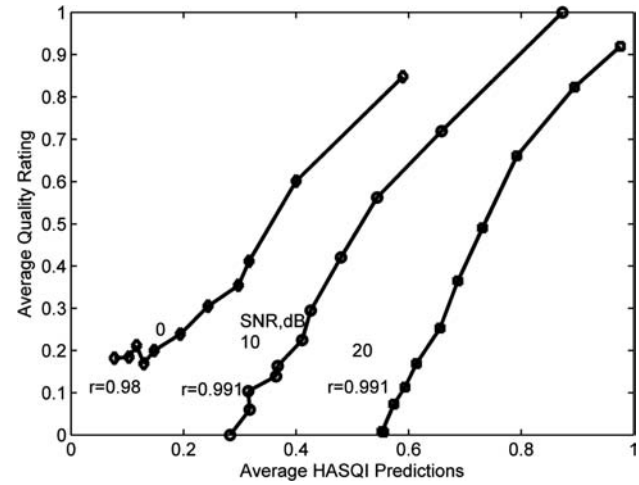

Fig. 11. Quality predictions for the modulated noise data, normal-hearing subjects.

The plots for the modulated noise dataset are presented in Fig. 11. The points at each SNR are linked. The ratings at each SNR are plotted as separate curves since the ratings were produced in separate test blocks, with each block having its own perceptual anchors of best and worst quality. The correlation coefficients between the HASQI predictions and the subject ratings are shown for each curve in the figure. The correlations are quite high, ranging from 0.991 for the SNRs of 20 and 10 dB to 0.980 for the SNR of 0 dB.

5 DISCUSSION

The accuracy of the quality index developed in this paper compares favorably to that of other indices in the literature. However, comparisons across different indices can be difficult due to several factors. First, different indices may be designed and optimized for different types of signal degradation and different amounts of degradation. Second, indices may be tested using different speech corpora and different languages. Third, indices may be based on

different assumed hearing-loss characteristics and reflect ratings from different subject groups. Finally, the subject ratings on which the indices are trained can vary for similar signal processing conditions due to the range and type of signal degradations included in each experiment.

The results in this paper compared nonlinear models that used differences in the temporal fine structure, the envelope, and a combination of the two. The temporal fine structure model was motivated by the nonlinear model of Tan et al. [1]. In Moore et al. [62], a combined model was formed using a weighted sum of the nonlinear model and a linear model that measured changes in the long-term signal spectrum. They obtained a correlation coefficient of 0.89 for sentence materials degraded by a combination of nonlinear distortion and linear filtering. When telephone transfer functions were added to the signals, the correlation coefficient was reduced to 0.85. The closest model in this paper is the combined model using the v^3 nonlinear term, which gave a correlation coefficient of 0.940 for the noise and distortion dataset. The improvement in performance may be due to the more complete auditory model used in HASQI. However, the c^2v version gives even better accuracy, with a correlation coefficient of 0.959 for the NH listeners.

A second quality index that incorporates changes in the temporal fine structure has been developed by Sen and Lu [63]. Their index uses a time-domain two-dimensional hydrodynamic model of the cochlea. The model incorporates the traveling wave along the cochlear partition, OHC motility, and IHC transduction. The speech stimuli included additive noise and low-data-rate speech coding without and with channel bit errors. The temporal signal changes that are measured are fast and slow temporal jitter, and spectral changes are also predicted. The temporal and spectral changes are combined using a second-order linear regression. They obtained a correlation coefficient of 0.881 using their model to predict the overall speech quality. The closest comparison in this paper is the noise and distortion dataset, for which the c^2v version of HASQI gave a correlation coefficient of 0.959.

The previous version of HASQI used cepstral correlation for the nonlinear term. Kressner et al. [64] used the previous version of HASQI to predict quality ratings from NH listeners for noisy speech that was processed through a variety of noise-suppression algorithms. They obtained a correlation coefficient of 0.86 for the original version of HASQI, and a correlation coefficient of 0.87 for a version that used a modified combination of the nonlinear and linear terms. The closest comparison in this paper is the noise suppression dataset, for which the old version of HASQI gave a correlation coefficient of 0.819 and the new c^2v version of HASQI improved the correlation coefficient to 0.891. If one knew a priori that only noise and noise suppression were present in the degraded signal, then even better performance could be obtained by using the c^3 version of HASQI, which gave a correlation coefficient of 0.932. However, one in general does not know the form of signal degradation in a hearing aid or communication system, and the c^2v version of HASQI has the advantage of giving good performance across a wide variety of conditions.

Suelzle et al. [65] applied the previous version of HASQI to speech quality ratings made by hearing-impaired listeners in rooms. Two rooms were used, one having a reverberation time of 0.1 s and one having a reverberation time of 0.9 s, and the stimuli were sentences in stationary speech-shaped noise or multi-talker babble. The HASQI values were computed separately for the signals recorded at the left and right ears of a KEMAR manikin and then averaged. They found that the HASQI algorithm returned values of zero for many conditions and used the raw cepstral correlation instead, which gave a correlation coefficient of 0.95. The closest comparison in this paper is for the noise and distortion dataset, which gives a correlation coefficient of 0.959 for the c^2v version of HASQI. The Suelzle et al. [65] results indicate that the HASQI modeling approach can be successfully extended to binaural listening in rooms even though the model was developed using data for monaural headphone listening.

Another quality index that compares the envelope of the degraded signal to the envelope of a clean reference signal is the PEMO-Q model of Huber and Kollmeier [8]. This model computes filters the signal envelope in each auditory band through a bank of modulation filters and then cross-correlates the output at each modulation frequency for the degraded signal with the output for the reference signal. The final value is a weighted average computed across modulation frequency and auditory analysis frequency. This model gave a correlation coefficient of 0.95 for speech processed through a variety of codecs. Rohdenburg et al. [66], also using PEMO-Q, found a correlation coefficient of just 0.70 for the overall quality of noisy speech processed through a noise-suppression algorithm and rated by NH listeners. A related experiment (Rohdenburg et al. [67]) returned a correlation coefficient of 0.81 for NH listeners. Pourmand et al. [68] found a correlation coefficient of 0.798 for PEMO-Q used to predict the quality of noisy speech processed through six different noise-reduction algorithms. The closest comparison in this paper is the noise suppression dataset for which the c^2v version of HASQI gave a correlation coefficient of 0.891 for the NH listeners.

The PESQ quality index (Beerends et al. [69]) is based on differences in the excitation patterns between the degraded and reference signals and, as such, is closest to the linear HASQI term. The performance results for PESQ (Beerends et al. [69]) give an average correlation coefficient of 0.935 for NH listeners evaluating a variety of telephone systems. Chen and Parsa [70] found correlation coefficients of 0.941 and 0.913 for PESQ applied to two different English-language telephone datasets rated by NH listeners and an average correlation coefficient of 0.864 for the results averaged over all of the languages and datasets that they evaluated. The closest comparison in this paper is the noise and distortion dataset, for which the c^2v version of HASQI gave a correlation coefficient of 0.959.

Rohdenburg et al. [66] and Rohdenburg et al. [67] found correlation coefficients of 0.81 when using PESQ to predict the benefit of noise suppression processing rated by NH listeners. Sen and Lu [63] found a correlation coefficient of 0.660 for PESQ used to predict the effects of

noise and low-data-rate speech coding for NH listeners. Hu and Loizou [71] found a correlation coefficient of 0.92 for PESQ used to predict the effects of noise and noise suppression algorithms for NH listeners, and Kressner et al. [64] measured a correlation coefficient of 0.91 for the same data. Pourmand et al. [68] found a correlation coefficient of 0.825 for PESQ (wideband version) used to predict the quality of noisy speech processed through six different noise-reduction algorithms. The closest comparison in this paper is the noise suppression dataset, for which the c^2v version of HASQI gave a correlation coefficient of 0.891 for the NH listeners.

The PESQ procedure has also been extended to hearing loss (Beerends et al. [72]). The modified version of PESQ was evaluated using speech having a variety of nonlinear distortions and linear filtering operations and presented over a telephone handset. HI listeners with losses ranging from mild to moderate rated the speech quality for Flemish or German materials, depending on the test site. The correlation coefficient for the modified PESQ, averaged over hearing loss and presentation language, was 0.913. In a subsequent experiment the level of presentation was varied over a +10 to –15 dB range and PESQ modified to account for the level of presentation; the correlation coefficient for this experiment was 0.92. The closest comparison in this paper is the noise and distortion dataset, which gave a correlation coefficient of 0.965 for the HI listeners.

The PESQ index has recently been extended to produce the POLQA quality index (Beerends et al. [73] [74]). The modified index includes estimates of the excitation pattern differences between the reference and degraded signals, the noise level, and amount of reverberation. The intermediate scores are converted to the final index value using a third-order polynomial mapping. For narrowband speech rated by NH listeners, the correlation coefficients ranged from 0.85 to 0.94; for wideband speech the correlation coefficients ranged from 0.91 to 0.96; and for super-wideband speech the correlation coefficients ranged from 0.90 to 0.94. The closest comparison in this paper is the noise and distortion dataset for which the c^2v version of HASQI gave a correlation coefficient of 0.959.

A combined modeling approach has also been proposed by Beerends et al. [75]. Their approach uses three models, one each for the effects of noise, linear filtering, and time-varying gain. The outputs of the three models are then combined using an L_p norm: the three values are raised to the 0.1 power, averaged, and then the average raised to the power of 10. For speech subjected to a combination of the three degradation mechanisms, they found a correlation coefficient for their model of 0.85 with ratings from NH listeners. Using the minimum of the three values rather than the L_p norm gave a correlation coefficient of 0.82, while the PESQ metric gave 0.79. Again, the closest comparison in this paper is the noise and distortion dataset, for which the c^2v version of HASQI gave a correlation coefficient of 0.959.

Another combined modeling approach has been proposed by Goldstein and Rix [76]. Their index is intended to predict the reduction in telephone speech quality due to both nonlinear effects, such as distortion due to a codec, and linear effects, such as the acoustic transfer function of a telephone handset. Several distortion features are extracted from the signal using an auditory model and combined using a neural network. Preliminary results give an overall correlation coefficient of 0.938 with quality judgments made by NH listeners for conditions involving acoustic, electric, and combined electric and acoustic signal processing. The PESQ standard gives a correlation coefficient of 0.918 for the same data set.

The final model considered in this set of comparisons is the E-Model of Möller et al. [77]. This model is intended for telephony and includes estimates of the noise level and SNR, echo, echo cancellation effects, the receiver frequency response, and the system delay. Results for the modified E-Model give a correlation coefficient of 0.86 with quality ratings made by NH listeners for noisy speech processed through handsets incorporating noise reduction, recorded, and presented to the listeners diotically over headphones.

The results summarized in this section show that HASQI version 2 compares favorably with other indices that have been developed. The new version is also more accurate than the previous version of HASQI when averaged over the range of processing conditions, and there were no noticeable weaknesses for any of the processing conditions considered. However, the accuracy of the index for specific forms of distortion depends on how the distortion is measured. If the specific type of signal modification is known a priori, then one could choose the model that gives the best accuracy for that type of distortion. For the noise suppression and modulated noise datasets, for example, the c^3 version is most accurate, while for the noise vocoder the v^3 version is most accurate. However, the version using the c^2v nonlinear term is the most accurate when the predictions are averaged over all of the datasets and is therefore recommended for the general problem of estimating quality when the distortion mechanisms are unknown.

A further consideration is that the envelope modulation in the HASQI cepstral correlation is lowpass filtered at 62.5 Hz. Many languages are characterized by envelope modulation frequencies below 20 to 30 Hz (Greenberg and Arai [78]; Souza and Rosen [79]), and the 62.5-Hz cutoff frequency used in HASQI produces accurate results for English sentences. However, quality indices developed for languages such as English may not be as accurate for tonal languages. Ebem et al. [80] evaluated speech quality for English and Igbo (a tonal language spoken in Nigeria) materials degraded by additive noise, telephone codecs, changes in level, and reduced bandwidth. They found that POLQA gave a correlation coefficient of 0.92 for English but only 0.78 for Igbo. Thus extending HASQI to other languages may require further investigation of the envelope-modulation cutoff frequency and the ratio of envelope to TFS information used in the model.

6 SUMMARY AND CONCLUSIONS

This paper has presented an improved index for predicting speech quality. HASQI compares the envelope and

TFS outputs of an auditory model for a degraded signal to the outputs of the model for a reference signal. The auditory models for the reference and for the degraded signal both incorporate the peripheral hearing loss. The auditory model includes the middle-ear transfer function, an auditory filterbank, outer hair-cell dynamic-range compression, two-tone suppression, and adaptation of the inner hair-cell firing rate. Hearing loss causes a shift in auditory threshold, broader auditory filters, a reduction in the dynamic-range compression ratio, and a reduction in the amount of two-tone suppression.

The index comprises a nonlinear term and a linear term. The nonlinear term includes the comparison of the reference and degraded signal envelopes using cepstral correlation. The short-time log spectrum is approximated using a set of half-cosine basis functions. The amplitude of each basis function fluctuates over time, and loss of quality is computed using the average of the cross-correlation of the basis functions for the reference and degraded signals. The comparison of the reference and degraded signal temporal behavior uses vibration correlation. The signals are divided into segments and the short-time coherence calculated for each segment in each frequency band. The coherence is weighted by the neural synchronization index as a function of frequency and averaged across time and frequency. The final nonlinear term is the product of the cepstral correlation squared with the vibration correlation.

The linear term is based on the long-term average outputs in each auditory analysis band. The standard deviation of the difference in the spectra of the reference and degraded signals is computed, along with the standard deviation of the differences in the spectral slopes. The linear term is a weighted sum of the two standard deviations. The final combined output of the quality index is then the product of the nonlinear and linear terms.

The new index has several advantages in comparison with previous indices. The index incorporates a model of the auditory periphery that is adjusted to reproduce the effects of hearing loss, so the resultant quality predictions are accurate for both hearing-impaired and normal-hearing listeners. As a result, the same quality index can be used for both normal-hearing and hearing-impaired listeners. The model combines changes in the signal envelope, temporal fine structure, and long-term spectrum, so it responds accurately to a wide variety of signal degradations rather than being targeted at just a narrow set of changes. In addition, many hearing aids contain nonlinear signal processing, such as dynamic-range compression or frequency compression, that is intended to improve the audibility of speech but which also introduces signal distortion. The new index accurately predicts the trade-offs between audibility and distortion that can occur for hearing-impaired listeners using these algorithms. Finally, the model is monotonic in all of its arguments, which ensures well-behaved extrapolation for signal changes that are worse than those used to train the index.

The reference signal for the hearing-impaired listeners is linear amplification using NAL-R compensation [27] for the hearing loss. This could be a potential limitation if some form of hearing-aid processing is ultimately found to yield superior quality. However, there was no processing in the training or validation data that yielded a significantly higher quality rating than NAL-R equalization. Thus NAL-R filtering appears to be a valid reference condition for the hearing-impaired listeners.

Another potential limitation of HASQI concerns the temporal alignment. The main application of HASQI is intended to be hearing aids rather than telecommunications systems. Thus aspects of digital speech encoding and transmission, such as gradual changes in the timebase over the course of a sentence, packet loss, and potential rapid realignment of the timebase following pauses, are not included in the model or in the provided temporal alignment. The cepstral correlation and short-time coherence calculations are not expected to be affected by small timing variations, but the new index needs to be evaluated for its sensitivity to timing effects and the accuracy of its predictions for digital telecommunications systems.

A third potential limitation of the new index is that HASQI was derived using data for monaural headphone listening. Additional research is needed to determine the accuracy of the index for real hearing aids, as opposed to simulated processing. Further research is also needed to deal with the acoustic effects of the head and ear that occur in real-world hearing-aid use, the effects of binaural listening, and the impact of room reverberation. An additional consideration is the relative importance of envelope and temporal fine structure for predicting quality for tonal languages.

A final consideration is differences in listening situations. Comparing results across experiments is difficult since each experiment involves different test conditions, which will generate different perceptual anchors. Thus identical stimuli in two different experiments could give rise to different listener ratings even though the HASQI scores are the same. HASQI was trained on the Arehart et al. [17] data, so a HASQI score relates directly to those data. However, HASQI should provide accurate relative ratings across a wide range of signal-processing and hearing-loss conditions.

7 ACKNOWLEDGMENTS

The research of author JMK was supported by a grant from GN ReSound. Author KHA was supported by a NIH grant (R01 DC60014) and by the grant from GN ReSound.

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THE AUTHORS

James M. Kates

Kathryn H. Arehart

James M. Kates received the Bachelor of Science and Master of Science degrees in electrical engineering from the Massachusetts Institute of Technology in 1971 and the professional degree of Electrical Engineer from M.I.T. in 1972. He retired in 2012 from hearing-aid manufacturer GN ReSound, where he held the position of Research Fellow. He is now Professor of Hearing Engineering Research Practice in the Department of Speech Language and Hearing Sciences at University of Colorado in Boulder. His research interest is signal processing for hearing aids, with a focus on predicting speech intelligibility and speech and music quality. He also teaches a graduate course in hearing-aid signal-processing concepts. He is a Fellow of the Acoustical Society of America and a Fellow of the Audio Engineering Society and is the author or co-author of 60 technical papers and holds 25 US patents. His book *Digital Hearing Aids* was published in 2008 by Plural Publishing, San Diego.

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Kathryn Arehart did her undergraduate studies at Stanford University, CA, and received her graduate training at the University of Washington, Seattle. Dr. Arehart is now professor in the Department of Speech, Language and Hearing Sciences at the University of Colorado at Boulder. She is also a certified audiologist. Her research focuses on the impact hearing loss has on listening in complex auditory environments, the evaluation of hearing-aid signal processing for speech-in-noise perception by persons with hearing loss, the perception of sound quality, and the interactive effects of aging, cognition, and hearing loss both on speech perception and on hearing aid outcomes.